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Roll No

EC-802 (GS)

B.E. VIII Semester Examination, June 2020

Grading System (GS)

Advanced Communication System

Time : Three Hours

Maximum Marks :70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Discuss the principle of multi carrier code division multiple access.
b) Explain the principle of single carrier modulation with frequency-domain equalization.
2. Write various properties of Maximal-length-sequences.
3. Describe the principle of OFDM and give implementation of transceivers.
4. What is CDMA? Explain. How does it employ spread spectrum techniques during the multiple access of the signals.
5. Describe the architecture of the cognitive transceiver.
6. In multi-mode relay selection cooperative scenarios, how the optimal relay to cooperate with is determined? When to cooperate with these relays? Explain with example.
7. Describe MIMO in detail.

OR

How spectrum sensing and spectrum sharing is done in cognitive radio? Explain in detail.

8. Write short notes on any two of the following:
 - i) Time hopping impulse radio.
 - ii) Multi user detection.
 - iii) Inter carrier interference.
 - iv) Principle of interweaving.
 - v) Collaborative networks.
